

I Began Hacking in 1958 and Then Survived Rock'n'Roll

HOW STEVIE WONDER, BOB MARLEY, JIMI HENDRIX &
LIZA MINNELLI PREPARED ME FOR INCIDENT RESPONSE
AND SECURITY

WINN SCHWARTAU, FRSA

SECURITY GUY SINCE 1983

WWW.WINNSCHWARTAU.COM

@WINNSCHWARTAU

My Dad. NBC. 1946.

Oscilloscope and Radar Dev. WWII



My Mom. 1943. NBC Mastering Engineer



My First DefCon



My Analogue Education

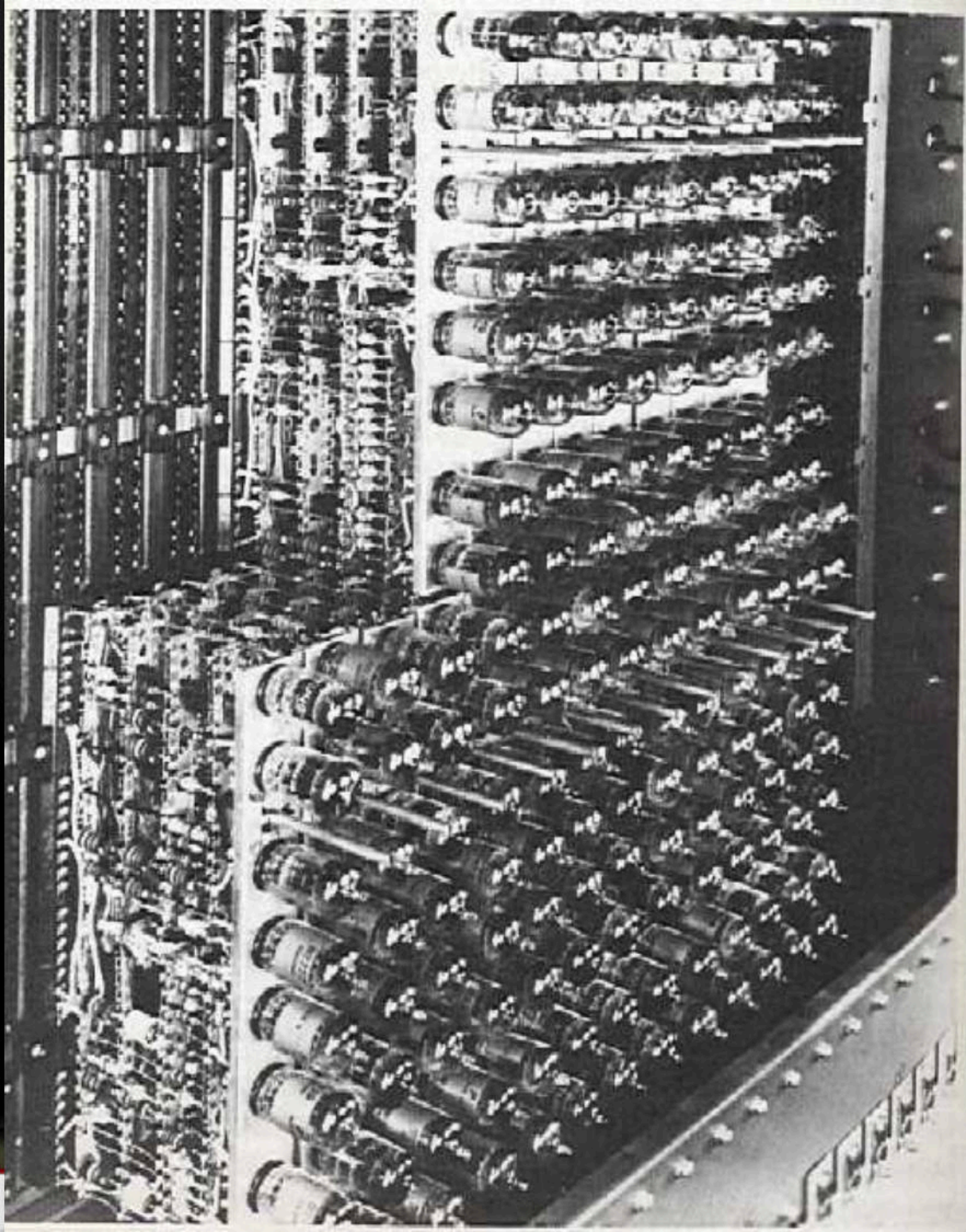


Winn As Young TV Repairman



My Electronics Store





Vacuum Tube Computer 1950s



My Home Phones 1950s: Some party line; Some 4-digit direct dial



1958: Phone Upgrade. My Mom Gets Pissed. Phone calls were *EXPENSIVE!*



New York Telephone

718 XXX-XXXX 240 R3A0

NOV 4, 1980

PAGE 2

DETAIL OF MONTHLY SERVICE AND EQUIPMENT

OUR POLICY IS TO PROVIDE YOU WITH AN ACCURATE MONTHLY BILL. PLEASE REVIEW THE FOLLOWING DETAIL OF MONTHLY CHARGES FOR SERVICE AND EQUIPMENT BILLED FOR THE PERIOD NOV 4 THRU DEC 3. THE TOTAL OF THE CHARGES DETAILED BELOW IS REFLECTED AS THE FIRST LINE OF CHARGES ON PAGE 1 OF THIS BILL.

***** THE FOLLOWING DETAIL IS NOT A BILL *****
IT IS PROVIDED ANNUALLY FOR YOUR REVIEW AND SHOULD BE RETAINED FOR YOUR RECORDS.

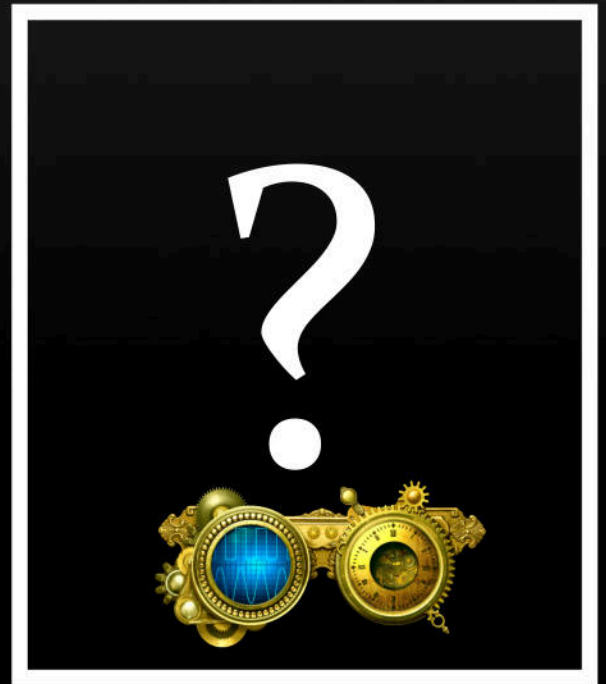
	PER MONTH
MAIN SERVICE LINE	6.04
1 WIRE OUTLET	1.24
1 TELEPHONE	1.18
CONTINUED	



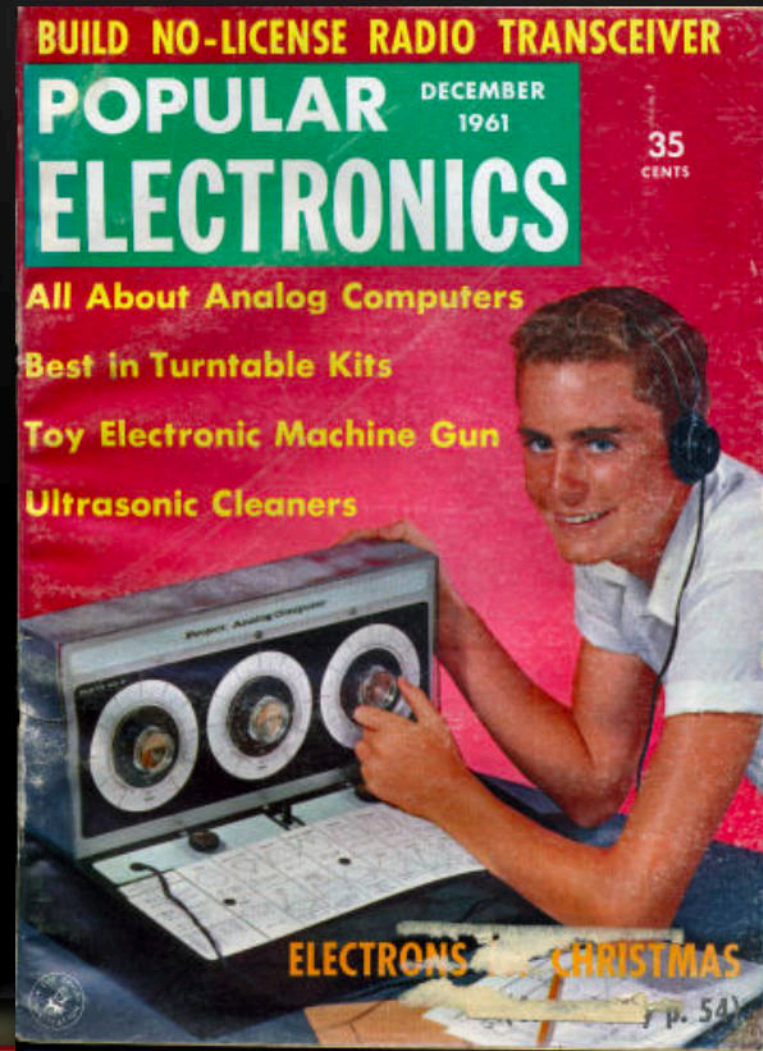
1958: So She Locked My Phone.



1958: How did I Make Phone Calls, Anyway?



1961



DIY – 1950-60s with Dickie Stallman



1962: Local Drug Store.
Worked There Part-time.
96th St. Columbus Ave., Manhattan



1962: Pay Phones



1962: Hack

Make Long Distance Calls for Free



A good buy—that's how your citizens will rate outdoor phones on public property. • Any hour, any day, outdoor phones let your citizens *save* time looking for a phone—*gain* the 24-hour convenience and peace of mind only a telephone can deliver. • Outdoor phones can mean extra revenue for your city's treasury, too, without adding one cent to your municipal tax burden. • Learn how a planned *network* of outdoor phones on public property can be an important part of your municipal planning. Just call your local Bell Telephone Manager and invite him to prepare a complete, no-obligation plan for your city.

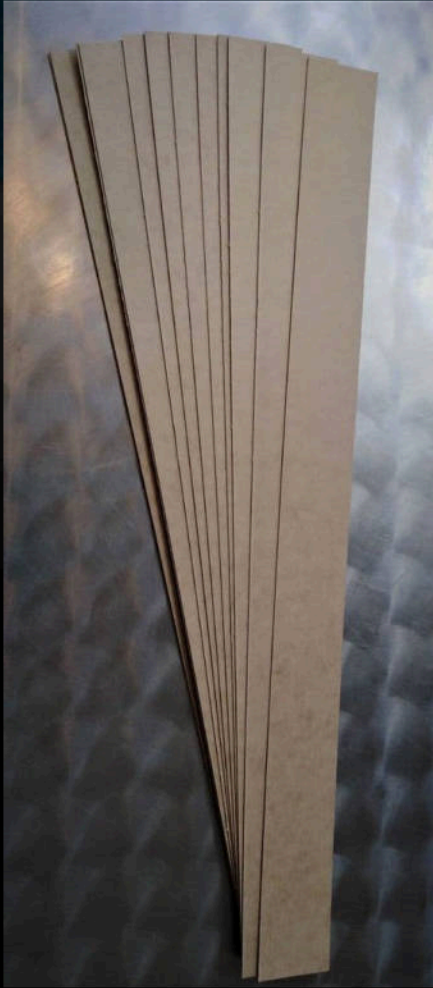


BELL TELEPHONE SYSTEM

- Goal
 - Free Local Calls
 - LD Phone Calls
 - Goof Calls



1962 Tools Used



High School Computer



Going Pro at 16: (1968)

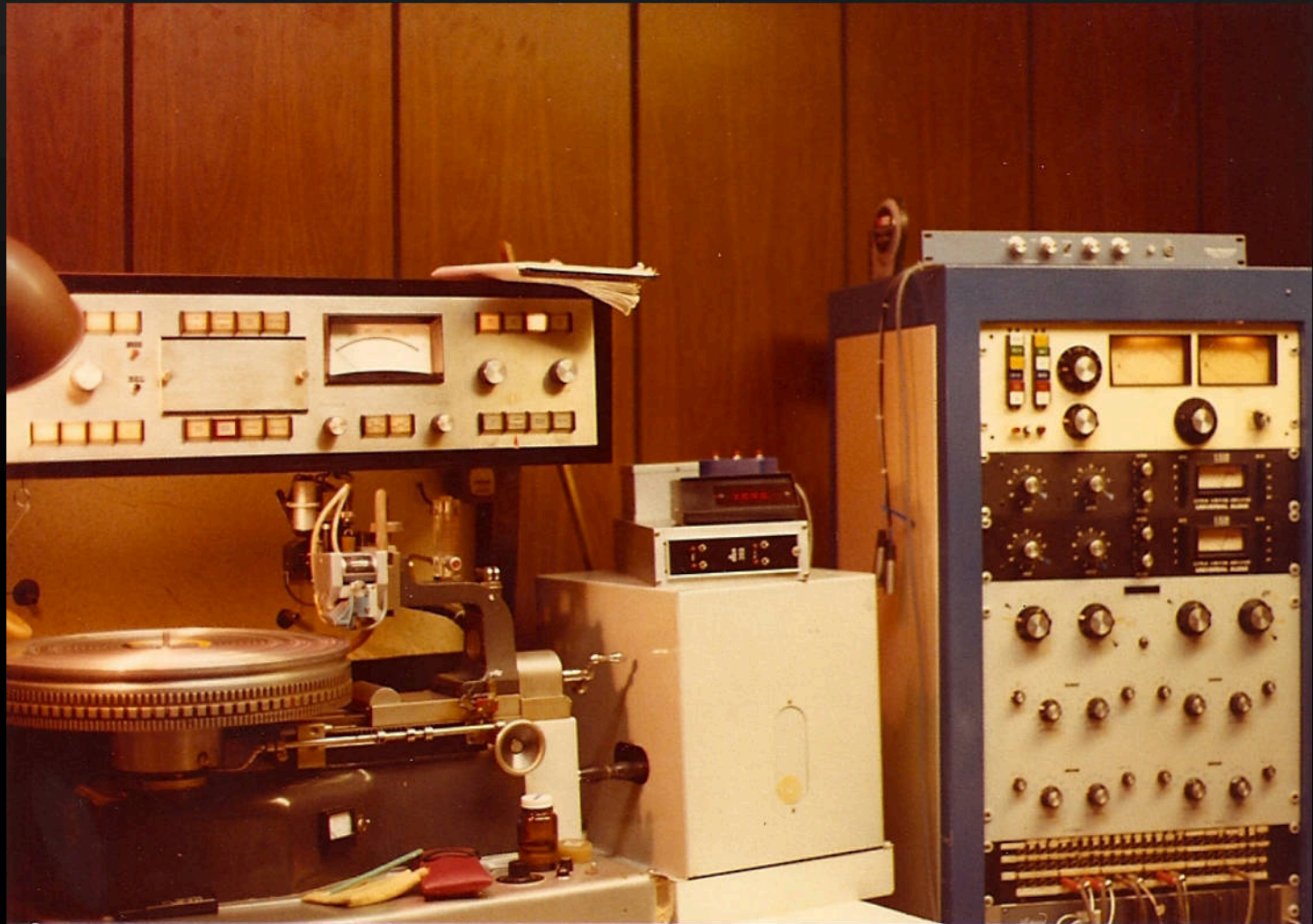
My First Studio Gig



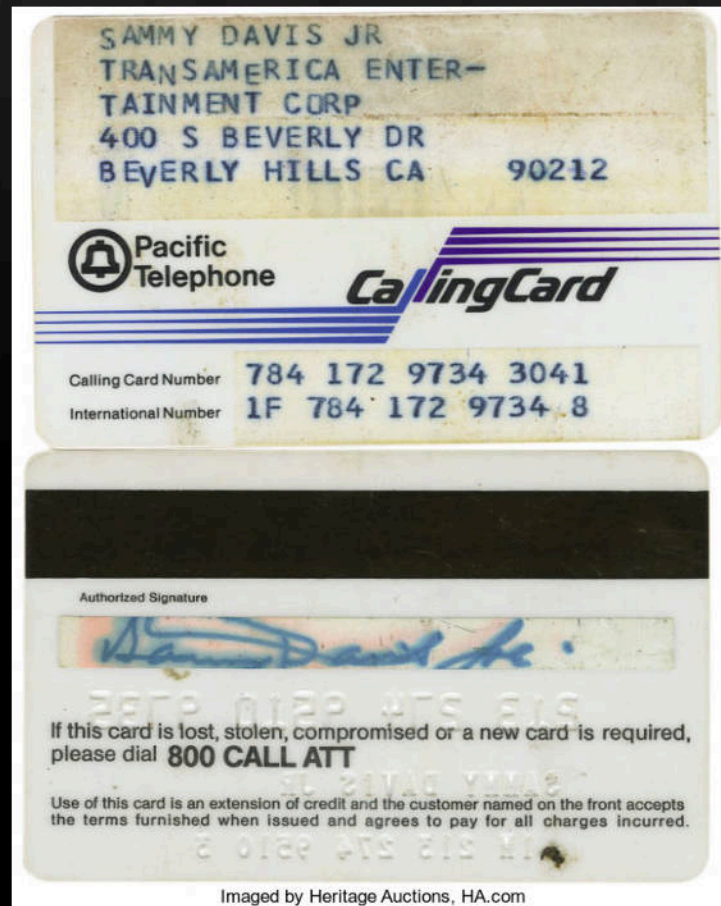
My First Sessions



My First Lathe



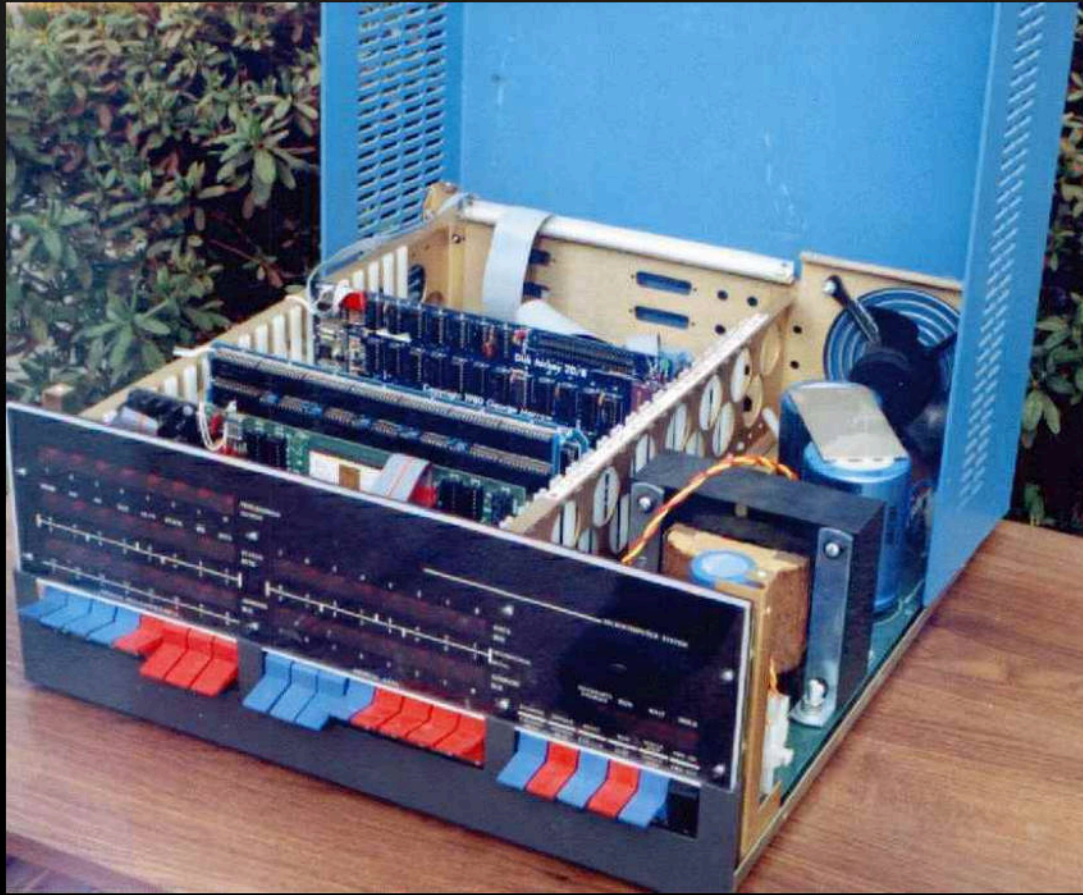
1969: Telephone Credit Card: Not a hack. Just theft.



1969-1970: Complex Systems



IMSAI 1972



Applause!

THE PALACE THEATRE

under the direction of Messrs. Nederlander

Joseph Kipness and Lawrence Kasha

in association with

Nederlander Productions and George M. Steinbrenner, III

present

LAUREN BACALL

in

APPLAUSE

Best Musical—1970 Tony Award

Book by **BETTY COMDEN** and **ADOLPH GREEN**

Music by **CHARLES STROUSE**

Lyrics by **LEE ADAMS**

(Based on the film "All About Eve" and the Original story by Mary Orr)

with

LEN CARIOU

GWYDA DONHOWE

LEE ROY REAMS

JOHN ANANIA

TOM URICH

ROBERT MANDAN

BRANDON MAGGART

BONNIE FRANKLIN

RAY BECKER

and

PENNY FULLER

Scenery by
ROBERT RANDOLPH

Musical Direction and
Vocal Arrangements by
DONALD PIPPIN

Production Associate
PHYLLIS DUKORE

Choreographic Assistant
TOM ROLLA

Costumes by
RAY AGHAYAN

Orchestrations by
PHILIP J. LANG

Lighting by
THARON MUSSER

Dance and Incidental Music
Arranged by
MEL MARVIN

Directorial Assistant
OTTO PIRCHNER

Production Stage Manager
TERENCE LITTLE

Directed and Choreographed by

RON FIELD

Original Cast Album by ABC Records

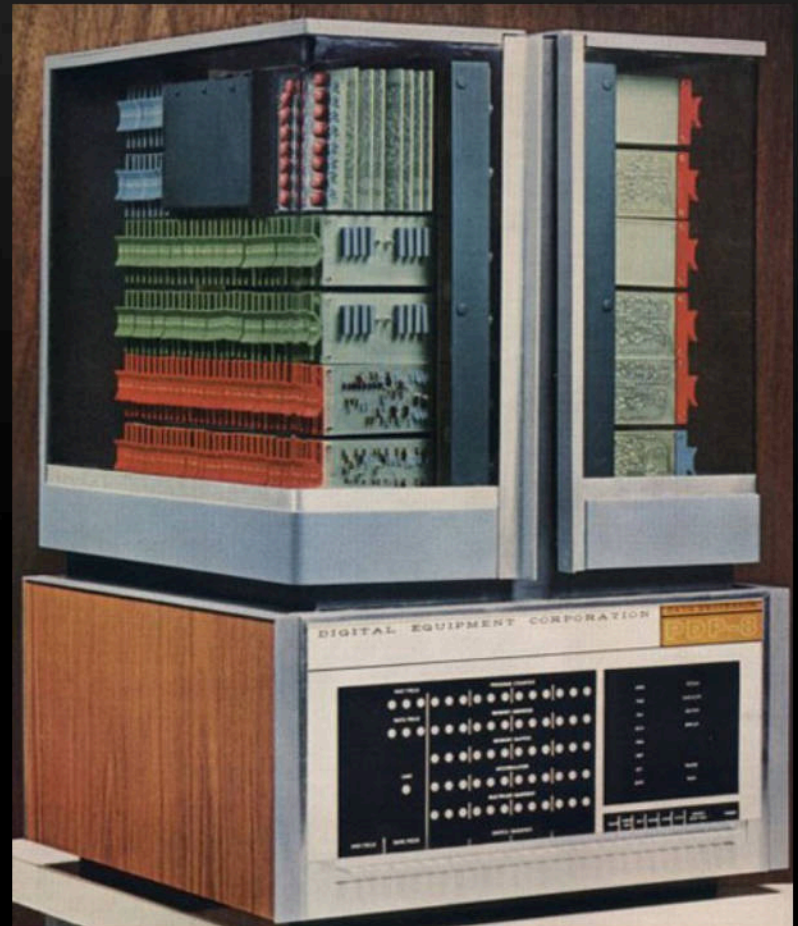
Music Publishers: Edwin H. Morris & Co.

The best seats cost more, too.

Old Grand-Dad
Head of the Bourbon Family



Kentucky straight Bourbon whiskeys. 86 proof and 100 proof bottled in bond. Old Grand-Dad Distillery Co., Frankfort, Ky. 40601.



1972: At the Studio



The Hack:

Using AT&T's leased equipment, we would run audio/music to the switchboard for internal distribution, tie lines to other locations, and even to California. Free.



We Were Broke: How Did We Hack Them?

TELEPHONE NUMBER	CC	RB	BO	RAC	DATE OF BILL	AMOUNT DUE
617 491 4937	962	0729	03	1	05 04 83	15 38

PLEASE SEND THIS CARD WITH YOUR PAYMENT

IF YOU PAY IN PERSON, BRING BOTH THIS CARD AND YOUR BILL



**New England
Telephone**

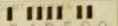
BOSTON, MASS. 02207

PLEASE DO NOT FOLD OR STAPLE THIS CARD

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34

DDM-21492

12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	
LOCATION										OPERATION										VARIABLE FIELD			

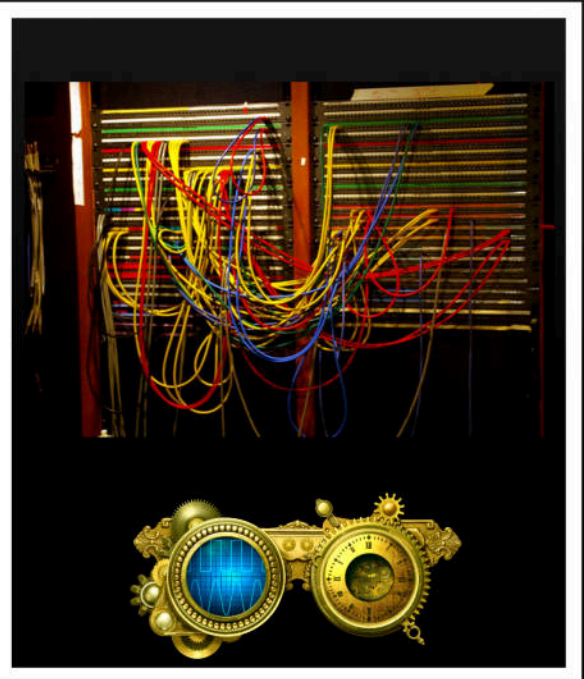
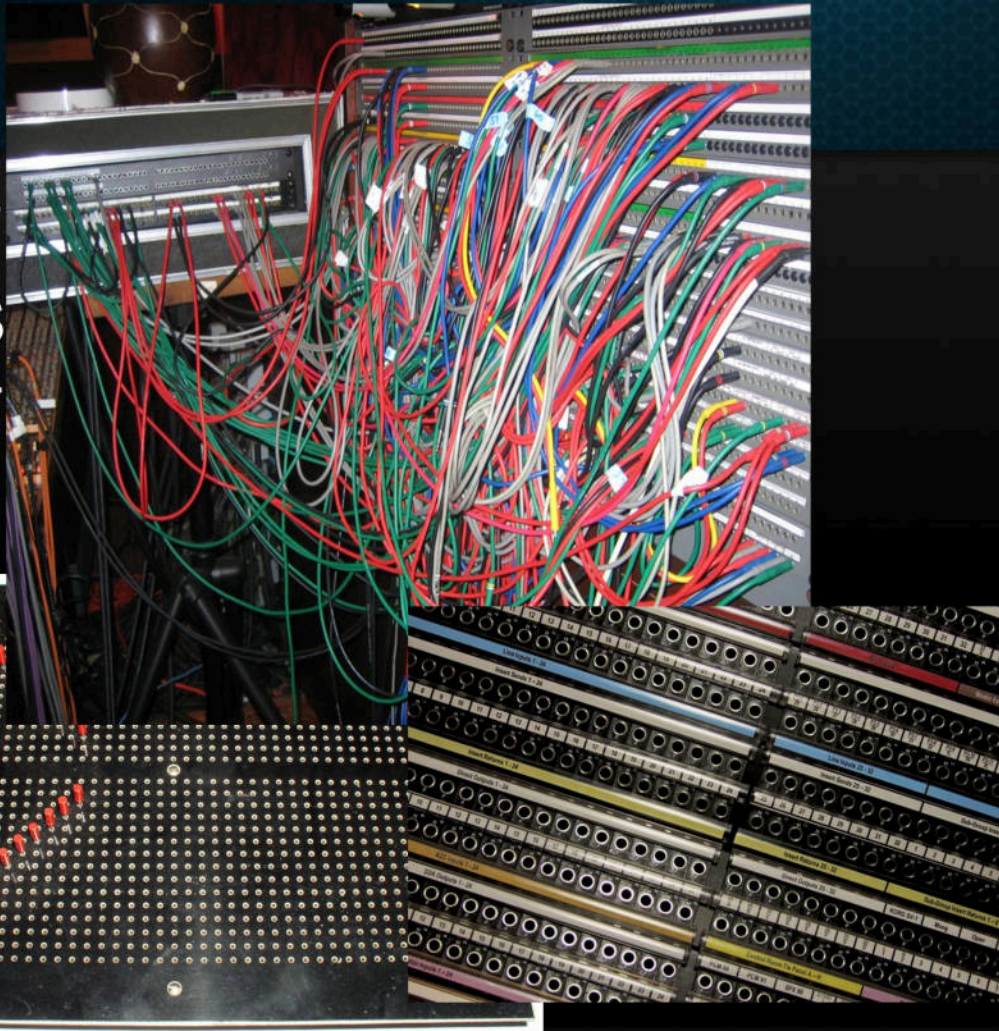
UNION ELECTRIC P. O. DRAWER 529 ST. LOUIS 3, MO		 PLEASE RETURN THIS STUB WITH PAYMENT	
RATE CODE		DUE BY	
APR 23 MAY 22 06 JUN 13 56		ACCOUNT NO. 	
METER READINGS		KILOWATT HOURS USED	
45824834		252	
AMOUNT OF BILL		DUE BY	
7.00		JUN 13	
AMOUNT OF BILL		7.00	

ALL ABOVE AMOUNTS INCLUDE 2% DISCOUNT SPTS. TAX WHERE APPLICABLE.
 HAVE INFORMATION MAY BE OBTAINED BY MAIL CONTACT SPTS.

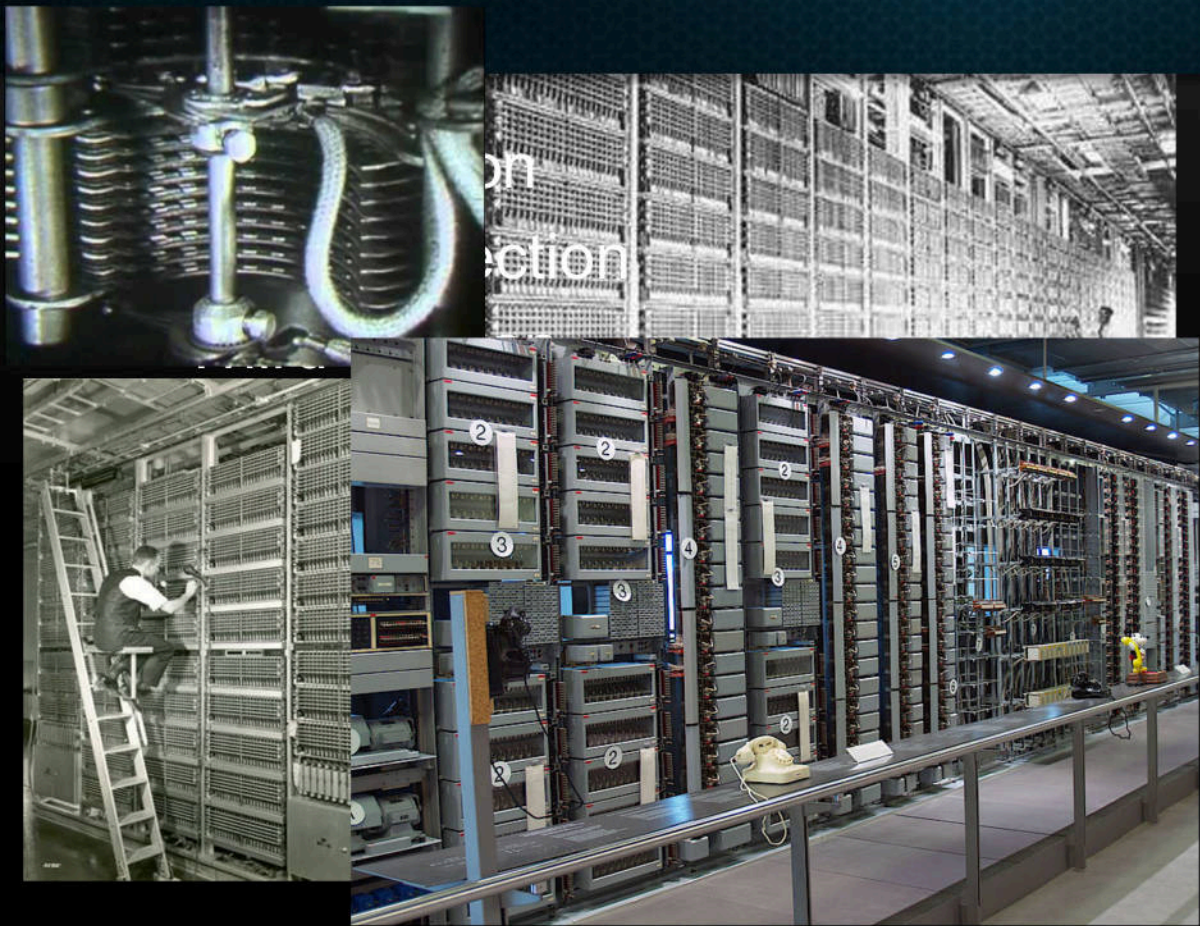
[illegible]

1975: They Wanted to build a Studio for Free.

- F
- S
- T



1975 Tools: AT&T Old Equipment Was Perfect!



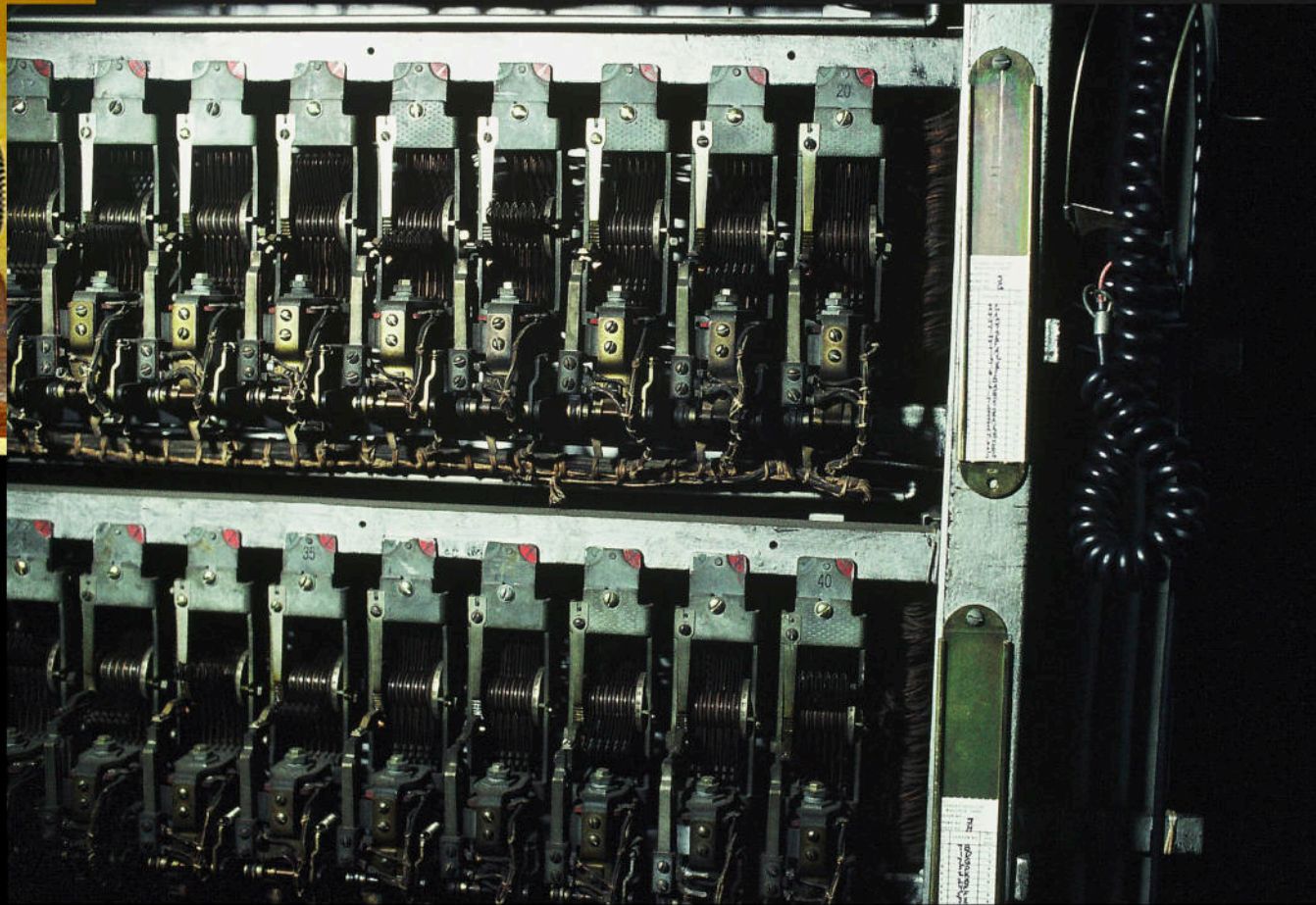
1970s: Dumpster Diving AT&T



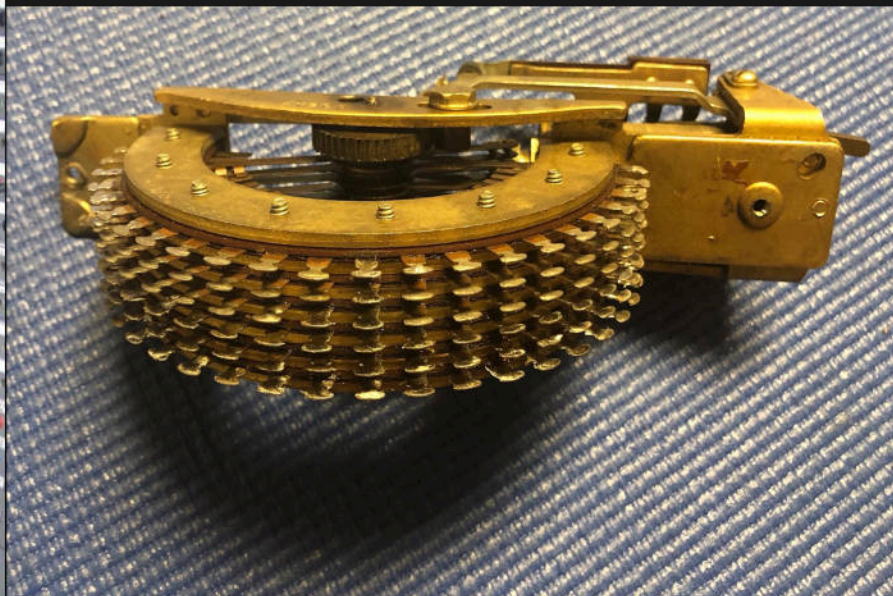
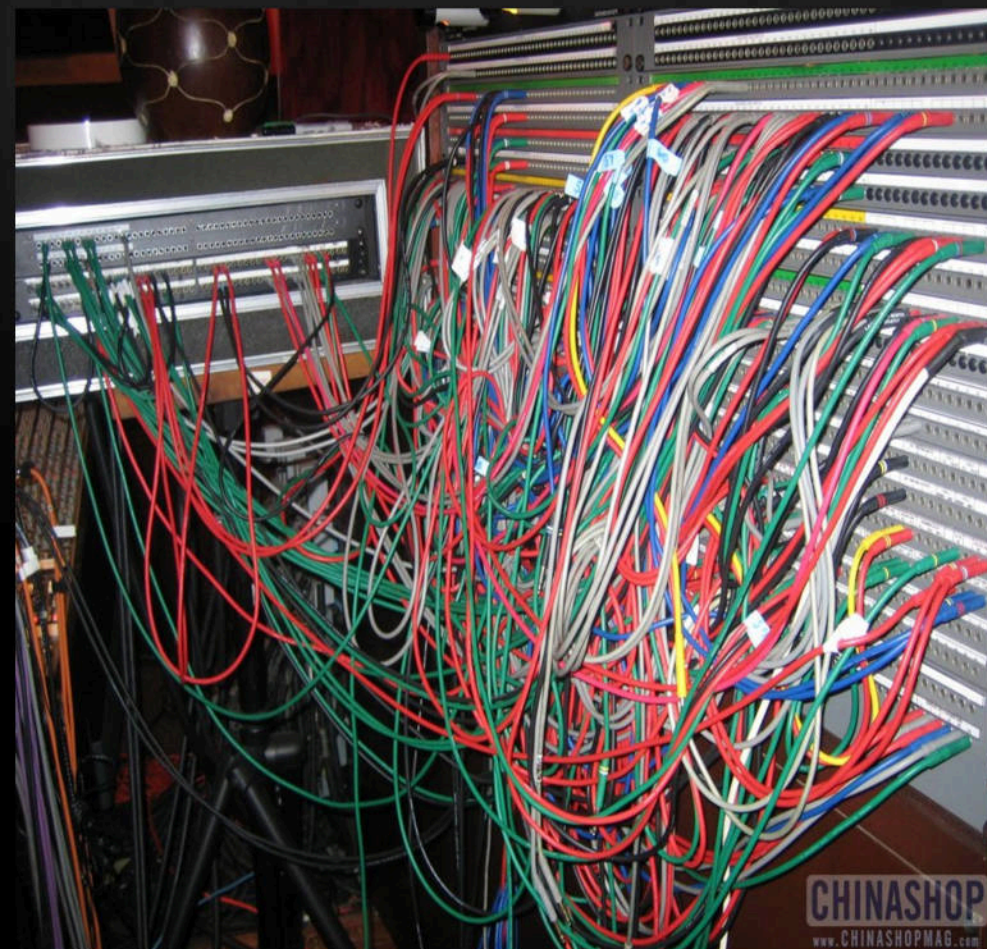
Find Great Repurposable Stuff!



Decade Counters for Studio Switching



Replaced This with These: Free



1978: My First Home Computer (That Worked At All)



My Wife. AES Convention 1980. First Computerized Audio



Automation

NEW Proprietary Transformerless Microphone Preamp is available at no extra cost in all 32C series consoles. Opening up a new world of sonic experience, it's a difference you can hear. Contact your Harrison distributor for a demonstration. For those who would prefer, the traditional transformer mike pre-amp continues to be available.

NEW 4832C Master Recording Console. 48 I/O module positions with VCA grouping and AUTOSET are ideal for mixing from two locked 24 track recorders. 48 I/O modules also allow effective split operation. Record on one group of modules and monitor on another group. Master modules may be placed in the center for easy access.

NEW 4432C Master Recording Console. A compact, light weight Harrison made especially for remote recording and any other weight or space sensitive application. Utilizing the standard 32C series modules, the Harrison 4432C does not include wood trim or patch bays. It does come with all normal patch points terminated in quick connect splice blocks for easy connection of full external patching. All "normalised" patch points are looped with jumpers on splice blocks so that full patching need not be installed for operation. A new compact aluminum frame and efficient wire routing keep weight extremely low for a full 44 X 32 console. The 4432C is ideal for studio applications where it is desired to customize patching and cabinetry.

NEW
FROM
Harrison

NEW High Resolution LED meters are now standard at no extra cost in the Harrison 32C series console. The Harrison high resolution meters have both true ASA VU Ballistics and true DIN PPM Ballistics available at the push of a button. In addition, each meter includes instantaneous peak detector to indicate overload of the recording medium. The Harrison high resolution meters have no digital clocks or high voltages to interfere with sensitive audio circuits. The 36-segment bar display gives a highly accurate and visible flicker-free display.

NEW AUTOSET Automation Programmer. A true multiprocessor system under software control, AUTOSET can store up to four separate and independent dynamic mixes on each channel of an audio recorder. An integral data cartridge can store up to 630 snapshot or preset mixes. Most important it's simple to operate. Even a guest mixer can use AUTOSET with only a couple of minutes instruction and an experienced operator can virtually perform miracles. Demonstrations can be arranged. Call today.

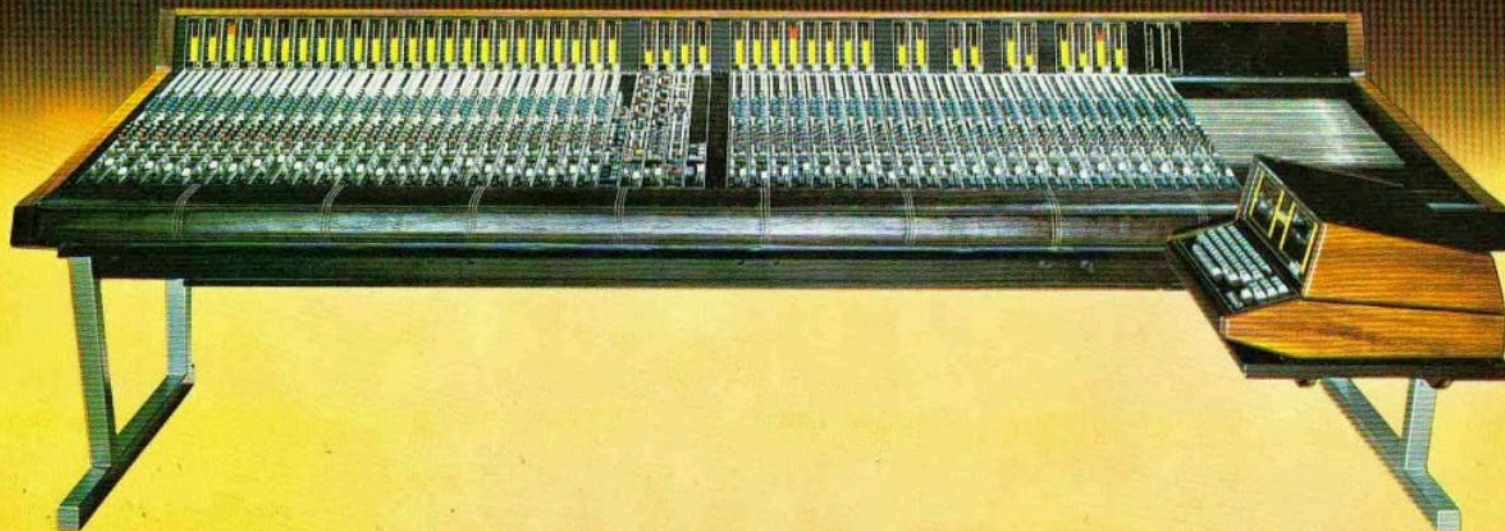
HARRISON PRODUCTS ARE AVAILABLE THROUGH THE FOLLOWING DISTRIBUTORS:

Factory
Harrison Systems, Inc.
Box 22964
Nashville, Tennessee 37202
(615) 834-1184 • Telex 555133
Dave Harrison • Tom Piper
Dave Purple

Eastern U.S.
Studio Supply Co.
P.O. Box 280
Nashville, Tennessee 37202
615-327-3075
Tom Kirby

International
Audio Systems International
128 N. Highland Avenue
Los Angeles, California 90036
213-933-2210 • Telex 686101
Paul Ford

Western U.S.
Electro Media Systems, Inc.
8230 Beverly Boulevard Suite 28
Los Angeles, California 90048
213-653-4931
Dan Gwynne



My Remote Live Studio: NYC 1980

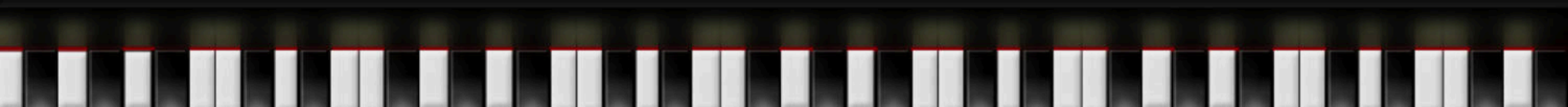


Incident Response Was a Way of Life

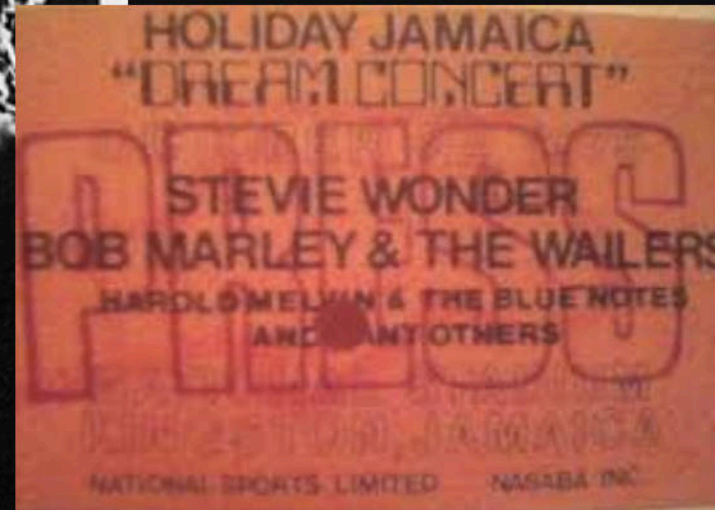
My younger self taught me, *“There is always a way!”*

The Show Must ALWAYS Go On
(Some EPIC Fails and Incident Response)

- # Stevie Wonder
- # Liza Minnelli
- # Charlie Daniels
- # Studio R-1
- # Lennon
- # Rod Stewart



Stevie Wonder & Bob Marley 1975 “Holiday Jamaica, Mon” (Free!)



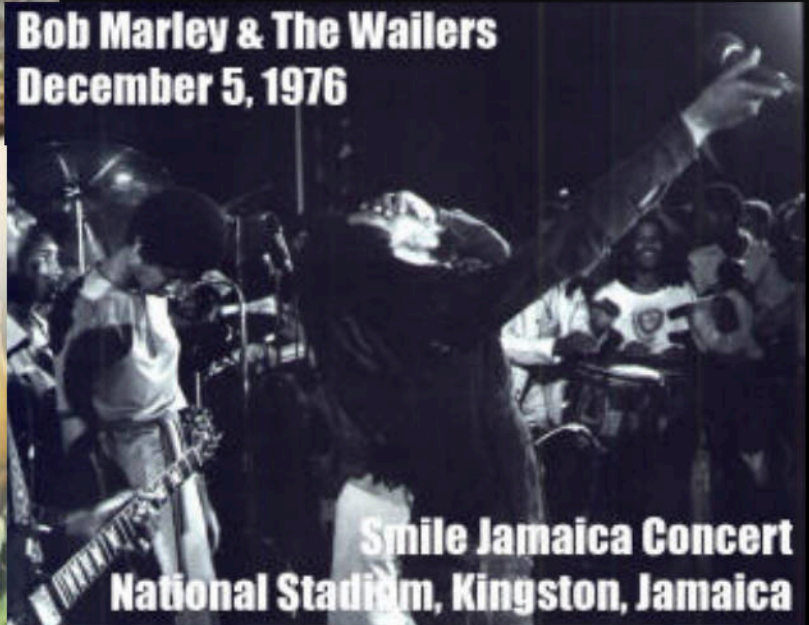
Stevie Wonder sits in with the Wailers, Kingston 1975.

Stevie Wonder Pt. I

USA Chaos: Lost 727



Bob Marley & The Wailers
December 5, 1976



Smile Jamaica Concert
National Stadium, Kingston, Jamaica



Stevie Wonder Pt. III

Whaddya Do?



Restoring Power: Only 2 of Us Arrested



Stevie Wonder

Security Take-Aways

- # Let the crazy guy try something crazy. What do you have to lose when 100,000 people start to riot?
- # Power is GOD!
- # Cyber is Physical.
- # Fast Incident Response is Critical!
- # Have a Backup. DR Plan. Exercise it. Again.
 - # Plan for double faults.
 - # Think Confocal Infrastructural Vulnerability
- # When the IT hits the fan...ask for forgiveness, not permission.



When the **IT Doesn't Sync



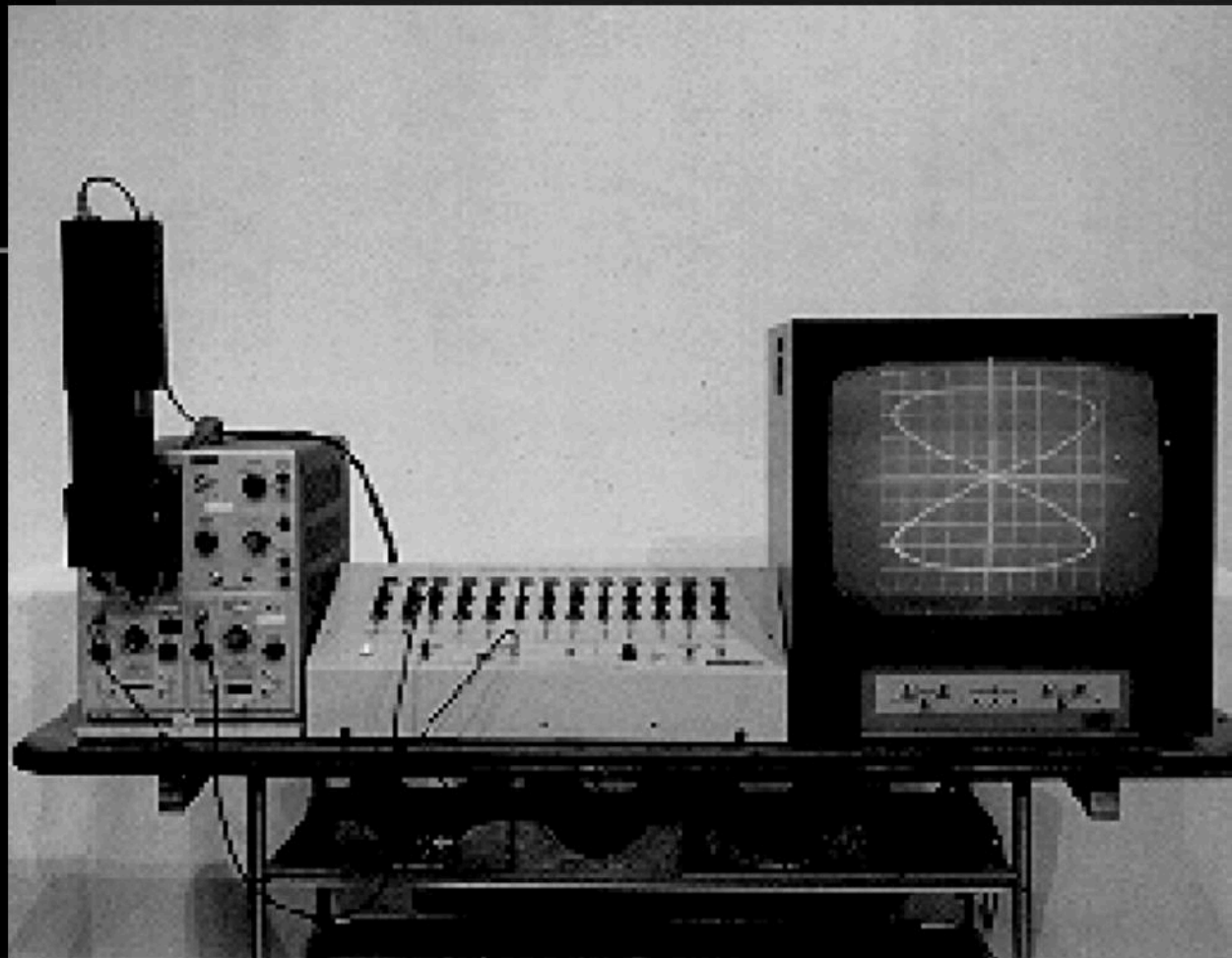
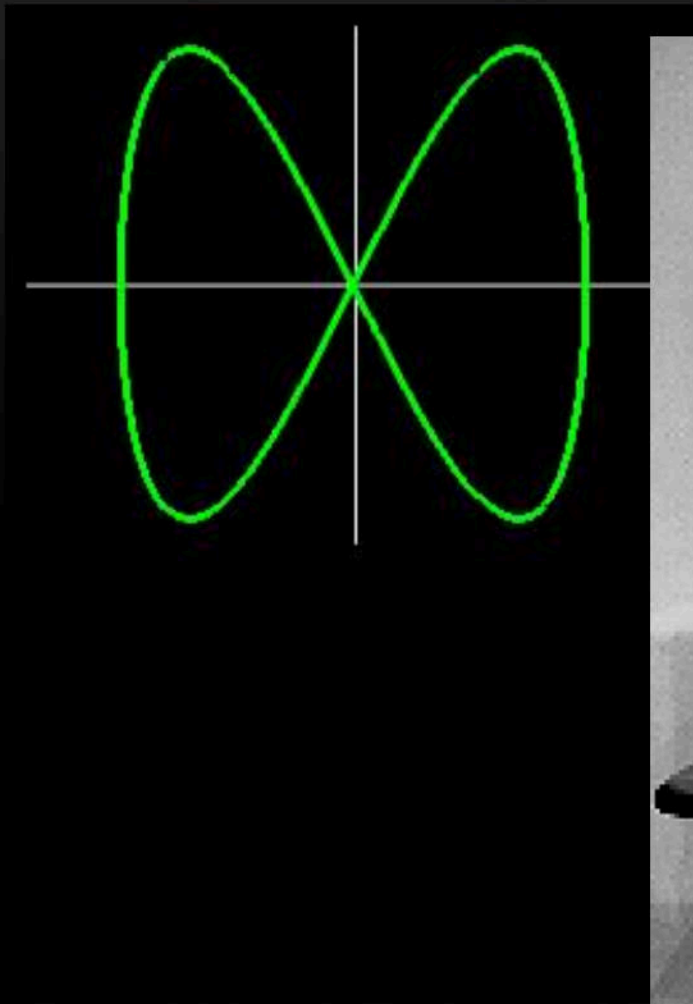
Live!



Auto Sync (Right!)



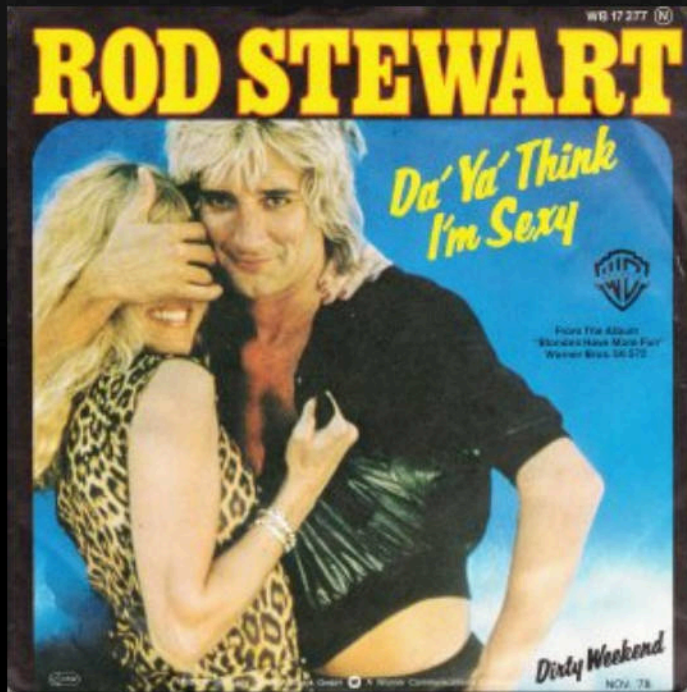
Manual Sync



More Sync Failures

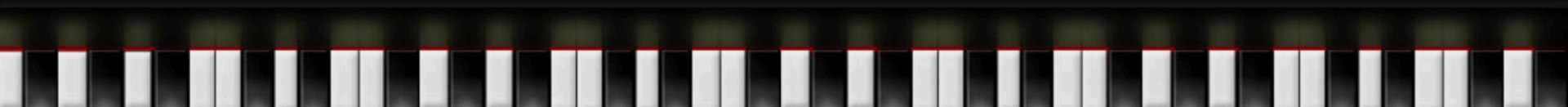


Only 8 Years Later...with Better Technology... Uh huh!



Liza Minnelli et al, Sync Failures: Security Take-Aways

- # Sync
 - # Policy, AD, AS, Mobile, Backup, DR, etc.
 - # Compliance, Policy, Testing?
 - # Cross NWs
 - # Time/Date-Event Stamps
(time changes globally!)
- # Can U Go Manual Synchronization Mode?
 - # Develop test & override
 - # Develop skills
 - # Have the manual tools in a kit
 - # Test the process
 - # Regularly



Charlie Daniels: MSG, NYC: 32:1 Collapse



Supposed to Be This



But Was Actually...



Charlie Daniels: Security Take-Aways

- # Plan for the Unimaginable
 - # Worst Case
 - # Trust No One
- # Architect for Graceful Degradation
 - # A Graceful Small Backup is Better Than None
- # Develop “What Can Go Wrongggggggg List”
 - # Risk/Cost Analysis
 - # Plan for Require Skills
- # Rehearse FAILURE Modes
 - # Mentally for those impossible situations
 - # Do it again

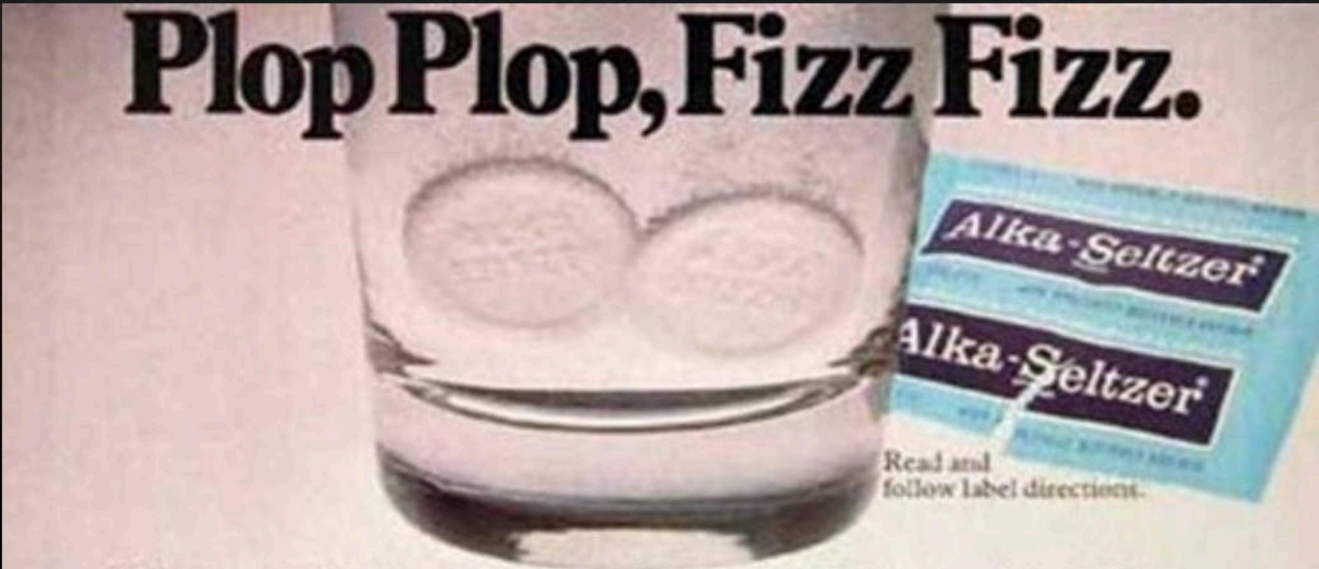


The Reality of Remotes = The Reality of Networks & Security

- # “You know, it’s always something.”
- # Murphy has a tent city here.
- # One backup is not enough.
- # Always, always, have Plan-B and Plan-C ready to go!
- # Test it all.
- # Overstaff.



Studio R-1: Flop Flop. Fizzle. Fizzle.



Plop Plop, Fizz Fizz.

Read and follow label directions.

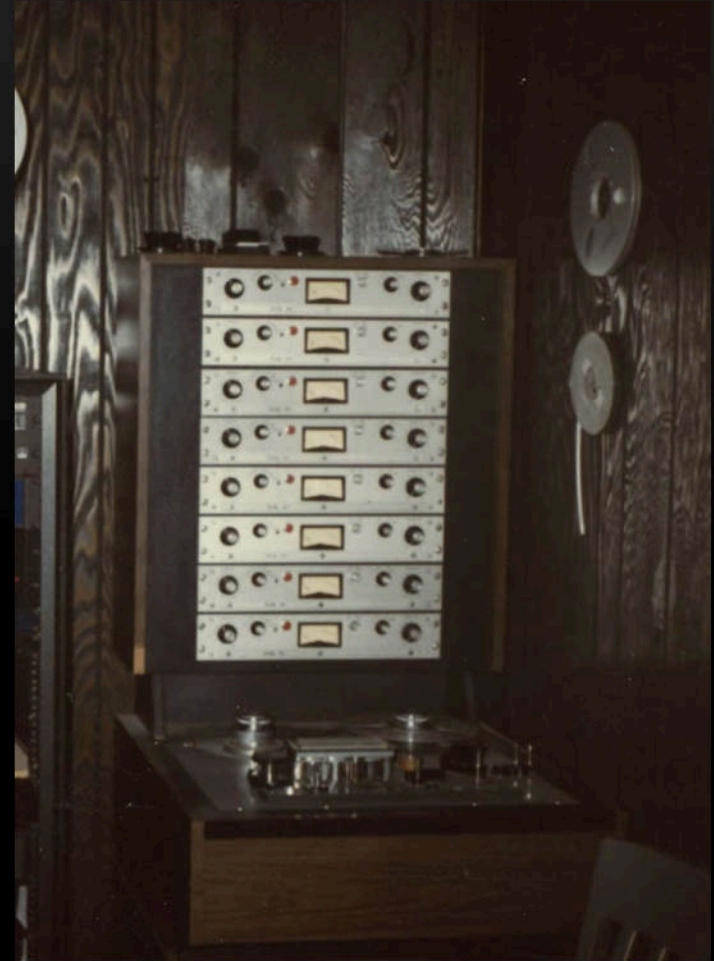
If there's one thing you always look forward to, it's a weekend party. You munch on chips and dip. You chug-a-lug your beer. You bugaloo till two. But sometimes you overdo it. You wake up feeling less than your best. When you do, reach for Alka-Seltzer. The moment you drink it, those tiny bubbles start to speed relief through your system. With specially buffered aspirin to soothe your throbbing head. And antacids to calm your upset stomach.

You'll be thankful you have Alka-Seltzer on hand. Because when morning comes, the only sound your aching head can bear to hear is a gentle plop plop, fizz fizz.

Alka-Seltzer
Oh, what a relief it is!

© 1994 Wm. S. Johnson & Co., Inc.

Supposed to Be...



What We Had To Do...



Studio R-1

Security Take-Aways

- # Time is Money: In the studio or the network!
- # Reliable, tested, fixed systems can fail, too!
- # Disaster Recovery
- # Graceful Degradation Part 2
- # Have a backup... Always!
- # Test the recovery process, not just the backup process.
- # Get very creative when the IT hits the fan.



Other Lessons in Systemic Collapse

1979:
Live From the
Lone Star Café
Hank Williams Jr.



1969:
Bill Haley and the
Comets
MSG NYC



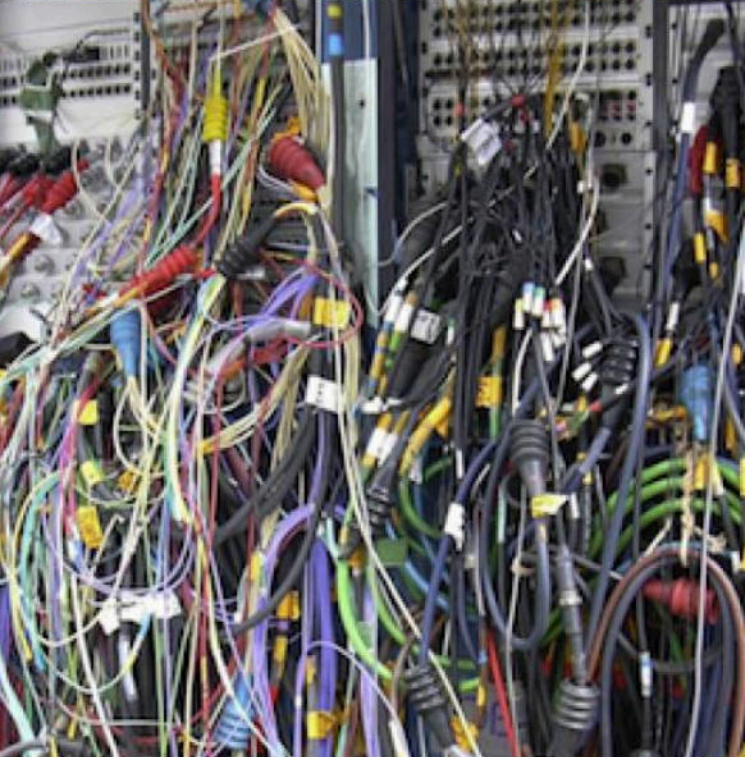
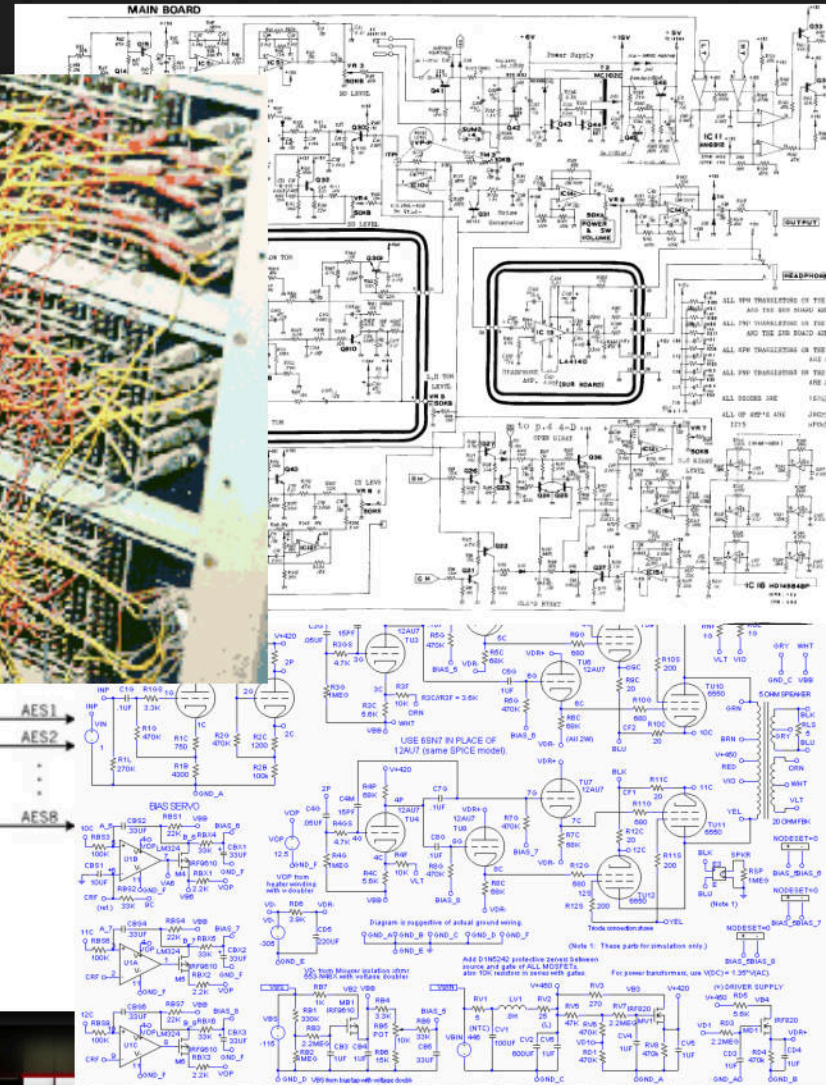
1977: BT Express
NYC Blackout



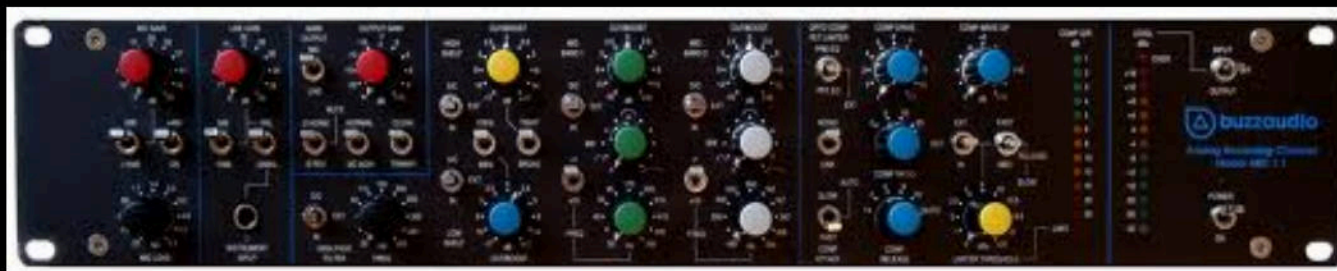
In the Studio, during a TV show, or on Live Sets...
you never knew what was going to go wrong.
But it did.

- # Massive Complex Networks
- # Hard to Visualize in One Go
- # It's a Way of Thinking
 - # Some call it Strategic
- # Easier to Spot Potentials for Failure
- # Complexity Breeds Insecurity

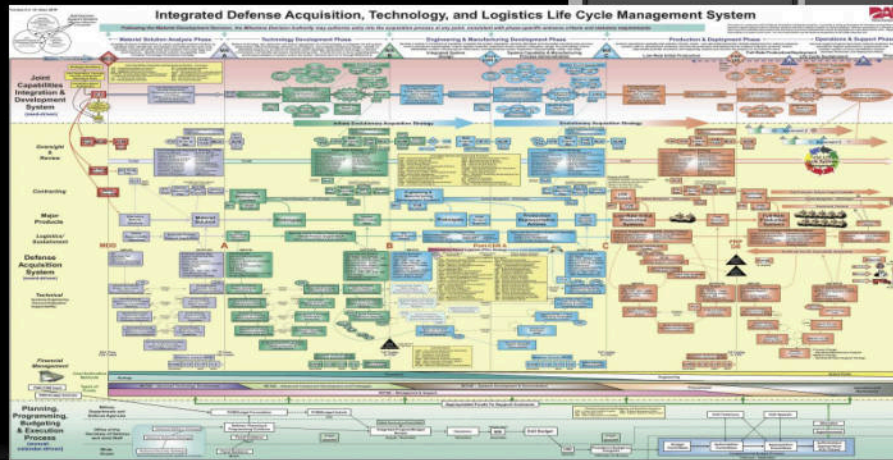
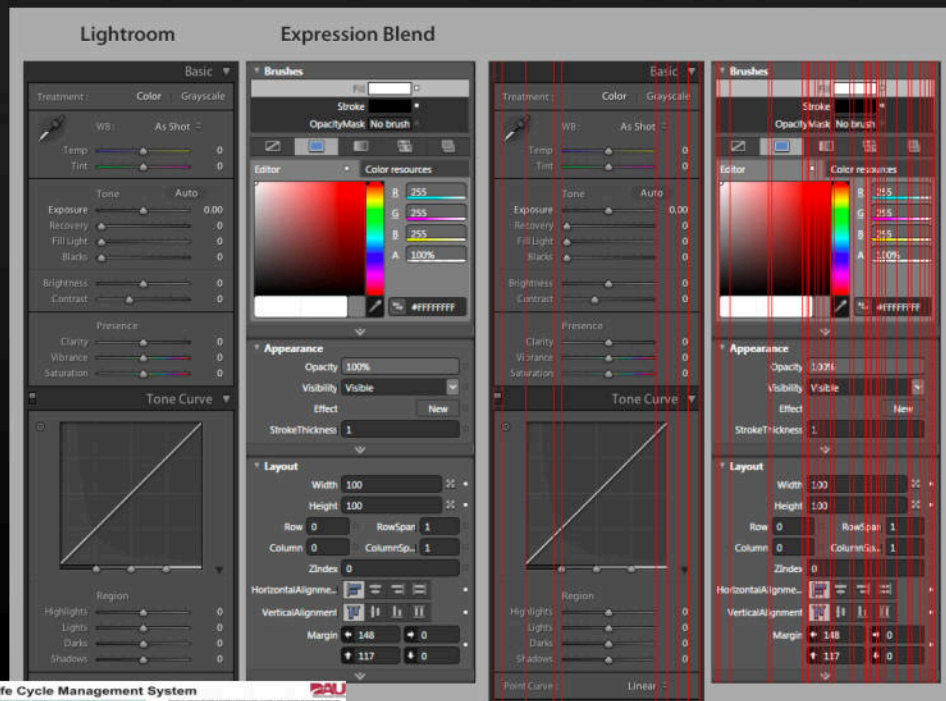
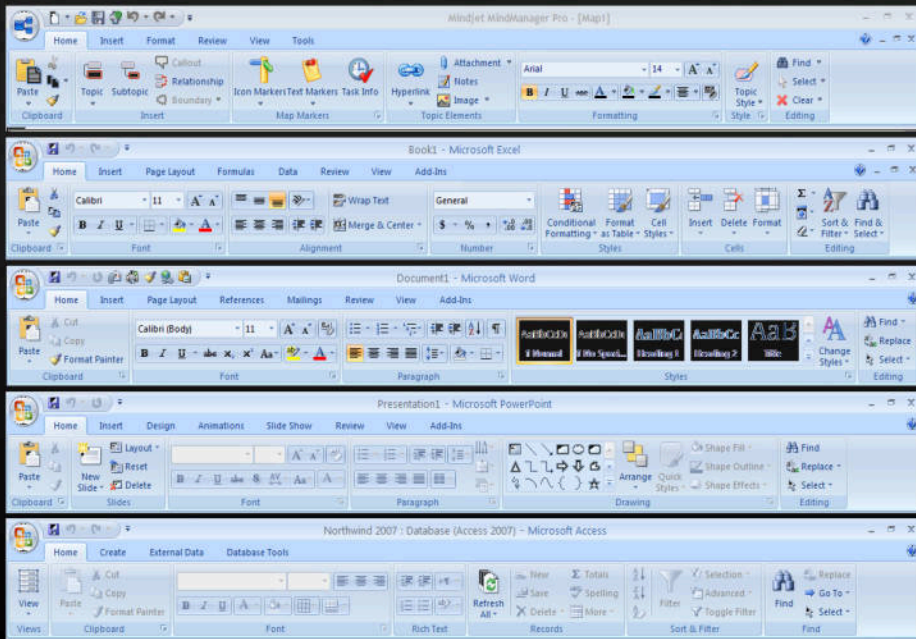




Turn All The Knobs



Turn All The V-Knobs



Final Incident Response Rant

- # Add Some Crazy
- # Teach Failure
- # Architect for Failure
- # Think Strategically
- # Learn Systems
- # Inter-Disciplinary Skills
- # More Hands-On Engineering
- # Practice Incident Response & Recovery Processes



Comments, Q & A?

Winn Schwartzau,
FRSA

www.WinnSchwartzau.com

[@WinnSchwartzau](https://www.instagram.com/WinnSchwartzau)

+1 727 393 6600

